

Backward error analysis and Bseries.jl

B-series

$$B(u, hf, y) := u(\emptyset)y + \sum_{\tau \in T} \frac{h^{|\tau|}}{\sigma(\tau)} u(\tau) F_f(\tau)(y),$$

In general, $u(\emptyset)$ may be 1 (for a map) or zero (for a flow)

The exact solution map: $y(t + h) = B(1/\gamma, hf, y)$

MATHEMATICS OF COMPUTATION, VOLUME 26, NUMBER 117, JANUARY, 1972

An Algebraic Theory of Integration Methods

By J. C. Butcher



B-series theory development

*In my opinion, at that time,
nobody in the world made the
necessary efforts to understand
Butcher's papers, except Ernst.
He then explained them to me,
and I tried to put them in a
more understandable form.*

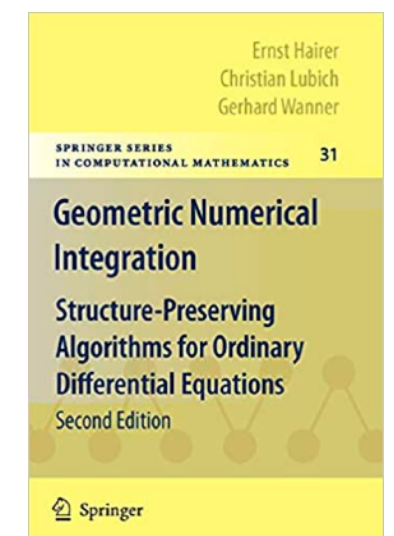
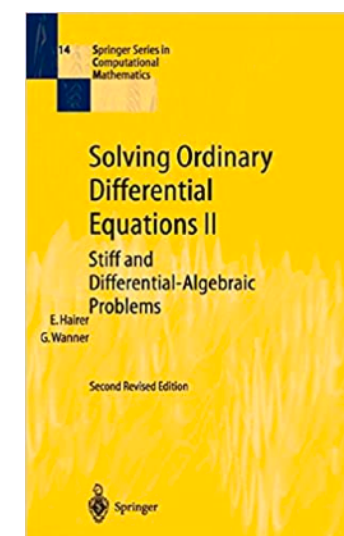
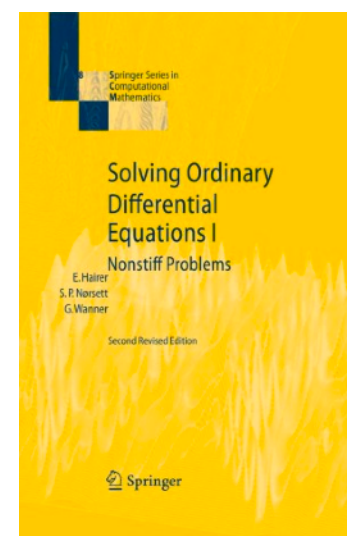
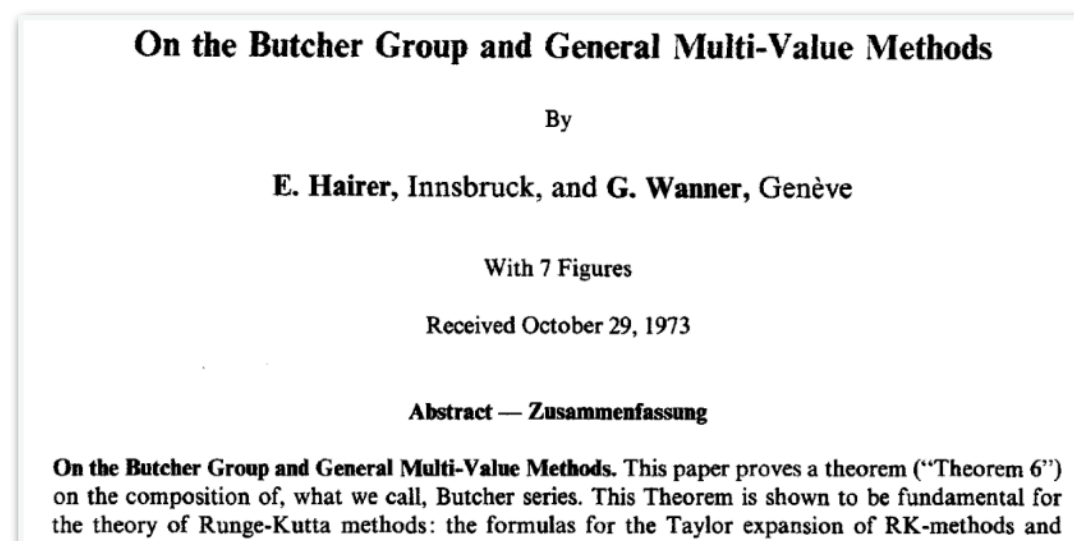
- Gerhard Wanner



Gerhard
Wanner



Ernst
Hairer



B-series (map) of an RK method

$$y_{n+1} = B(\Phi, hf, y_n)$$

$$\Phi(\tau) = \sum_i b_i \Phi_i(\tau)$$

τ	$ \tau $	$\sigma(\tau)$	$\gamma(\tau)$	$F(\tau)$	$\Phi_j(t)$
	1	1	1	f^j	1
	2	1	2	$f_k^j f^k$	a_{jk}
	3	2	3	$f_{kl}^j f^k f^l$	$a_{jk} a_{jl}$
	3	1	6	$f_k^j f_l^k f^l$	$a_{jk} a_{kl}$
	4	6	4	$f_{klm}^j f^k f^l f^m$	$a_{jk} a_{jl} a_{jm}$
	4	1	8	$f_{km}^j f_l^k f^l f^m$	$a_{jk} a_{jm} a_{kl}$
	4	2	12	$f_k^j f_{lm}^k f^l f^m$	$a_{jk} a_{kl} a_{km}$
	4	1	24	$f_k^j f_l^k f_m^l f^m$	$a_{jk} a_{kl} a_{lm}$

Simple examples

Explicit Euler method: $y_{n+1} = y_n + hf(y_n)$

$$y_{n+1} = y_n + hF(\bullet)$$

Implicit Euler method: $y_{n+1} = y_n + hf(y_{n+1})$

$$y_{n+1} = y_n + hF(\bullet) + h^2 F(\bullet\bullet) + h^3 \left(F(\bullet\bullet\bullet) + \frac{1}{2} F(\bullet\blacktriangledown\bullet) \right) + \dots$$

Example:

2-stage, 2nd-order RK

0	0	
$\frac{1}{2\alpha}$	$\frac{1}{2\alpha}$	0
	$1 - \alpha$	α

$$\begin{aligned}
 y_{n+1} = y_n &+ hF_f(\bullet)(y_n) + \frac{1}{2}h^2F_f(\bullet\!\!\!\begin{array}{c} \bullet \\ \vdots \\ \bullet \end{array})(y_n) + \frac{1}{8\alpha}h^3F_f(\bullet\!\!\!\begin{array}{c} \bullet \\ \diagdown \\ \bullet \end{array}\!\!\!\begin{array}{c} \bullet \\ \diagup \\ \bullet \end{array})(y_n) \\
 &+ \frac{1}{48\alpha^2}h^4F_f(\bullet\!\!\!\begin{array}{c} \bullet \\ \diagdown \\ \bullet \end{array}\!\!\!\begin{array}{c} \bullet \\ \diagup \\ \bullet \end{array}\!\!\!\begin{array}{c} \bullet \\ \diagdown \\ \bullet \end{array})(y_n) + \frac{1}{384\alpha^3}h^5F_f(\bullet\!\!\!\begin{array}{c} \bullet \\ \diagdown \\ \bullet \end{array}\!\!\!\begin{array}{c} \bullet \\ \diagup \\ \bullet \end{array}\!\!\!\begin{array}{c} \bullet \\ \diagdown \\ \bullet \end{array}\!\!\!\begin{array}{c} \bullet \\ \diagup \\ \bullet \end{array})(y_n) + O(h^6)
 \end{aligned}$$

What can you do with B-series?

- Order conditions
- Method composition
- Backward error analysis (modified equations)
- Modifying integrators
- Analysis of method properties
- ...

Advanced computation with B-series

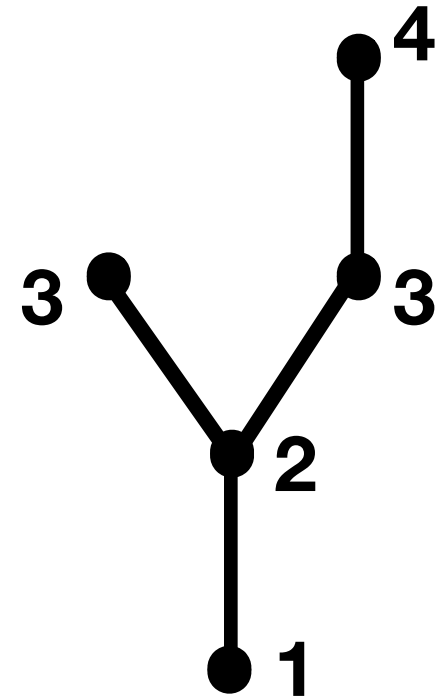
- Finding modified equations or modifying integrators requires extensive calculations (both for the coefficients and the elementary differentials) and is error-prone
- Doing this by hand is infeasible except for very low orders
- Using a computer algebra system doesn't immediately alleviate this, due to the need to manipulate subtrees and forests
- Solution: fully automate the calculations

bseries.jl

- Written in Julia
- Performance optimizations make it possible to work with B-series of high order (~10)
- Also computes the elementary differentials for concrete ODE systems
- <https://github.com/ranocha/bseries.jl>
- Order conditions 
- Method composition 
- Backward error analysis (modified equations) 
- Modifying integrators 

Trees as level sequences

- Label each node by its distance from the root
- Traverse the tree depth-first
- Canonical representation: use the lexicographically greatest sequence



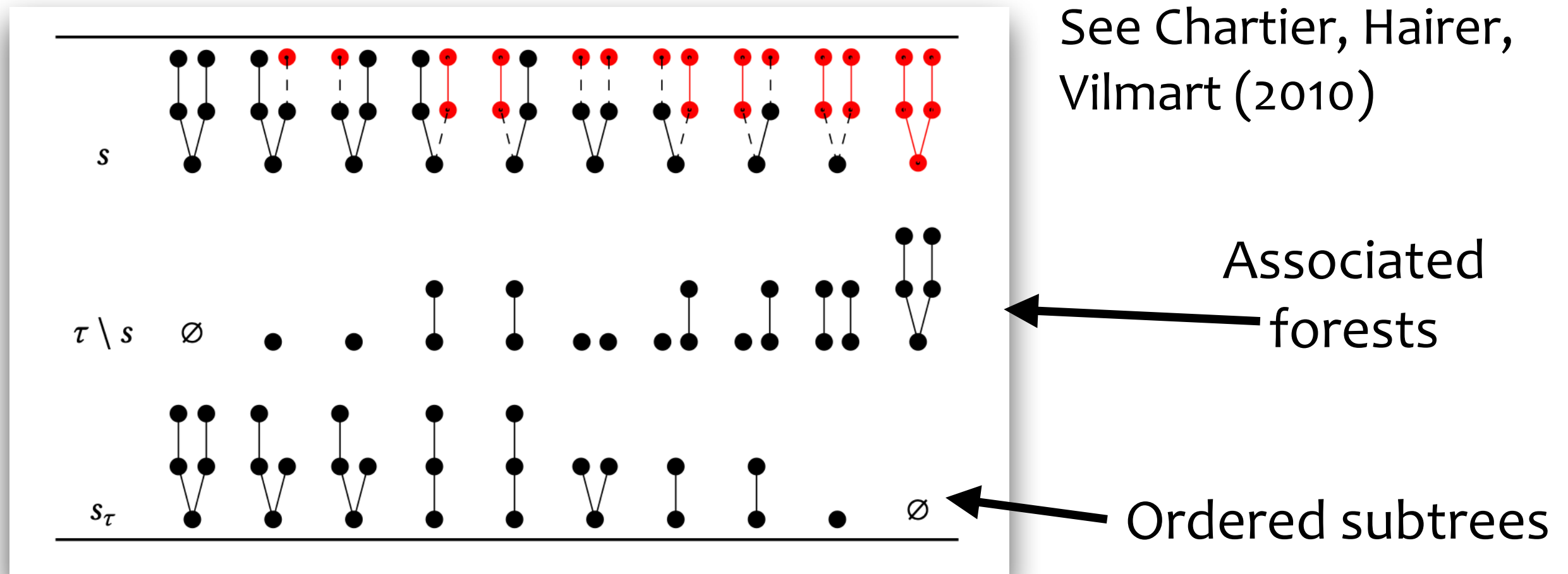
[1,2,3,3,4]

[1,2,3,4,3]

The B-series of the composition
is the composition of the B-series

$$B(u, hf, B(v, hf, y)) = B(v \cdot u, hf, y)$$

$$(v \cdot u)(\tau) = \sum_{s \in S(\tau)} v(\tau \setminus s) u(s_\tau)$$



Composition Examples

- Is it possible to construct a 3rd-order method by composing two different RK22 methods?

$$\begin{array}{c|c} 0 & \\ \frac{1}{2\alpha_1} & \frac{1}{2\alpha_1} \\ \hline & 1 - \alpha_1 \quad \alpha_1 \end{array}$$

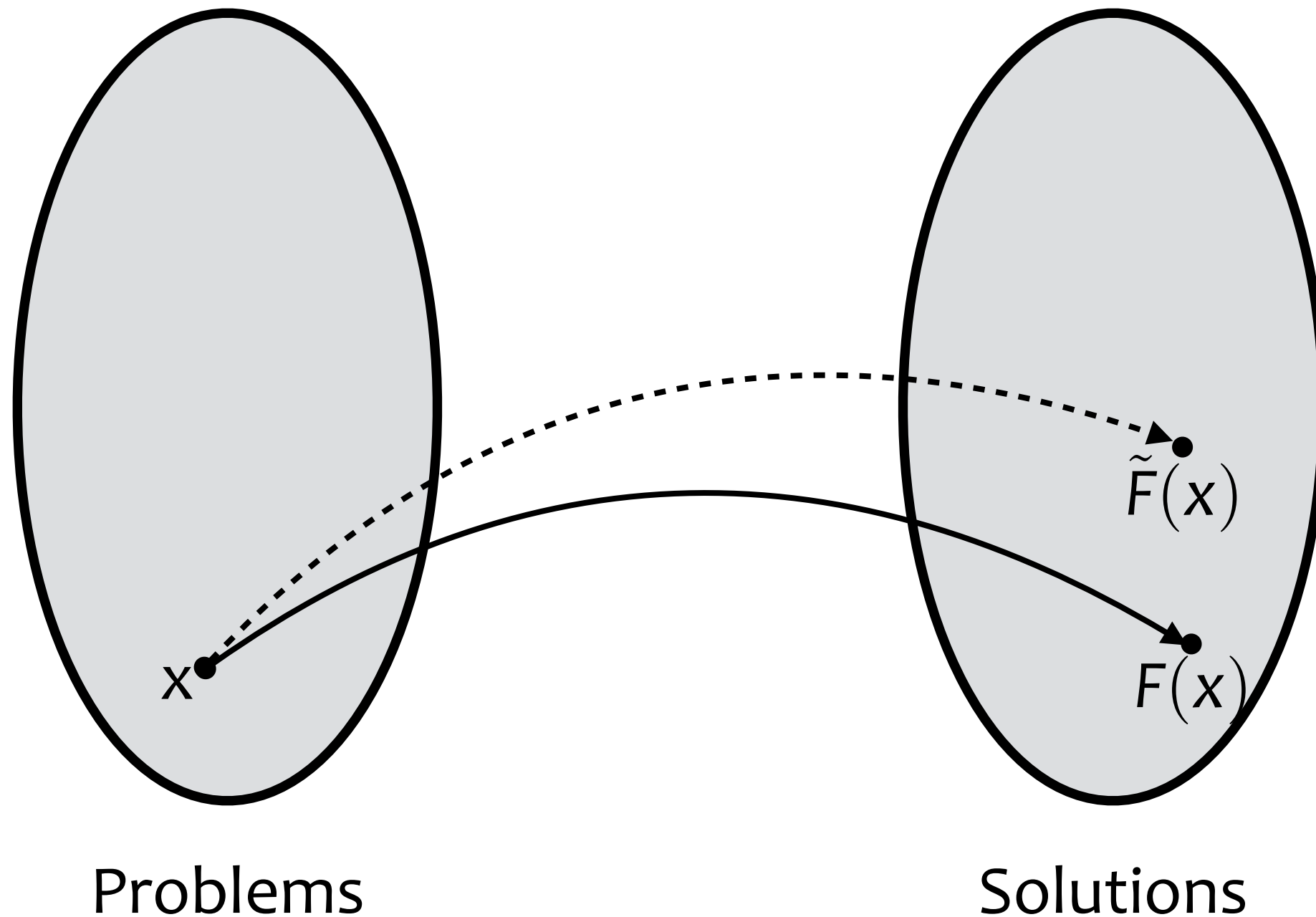
$$\begin{array}{c|c} 0 & \\ \frac{1}{2\alpha_2} & \frac{1}{2\alpha_2} \\ \hline & 1 - \alpha_2 \quad \alpha_2 \end{array}$$

`comp = compose(series1, series2, normalize_stepsize=true)`

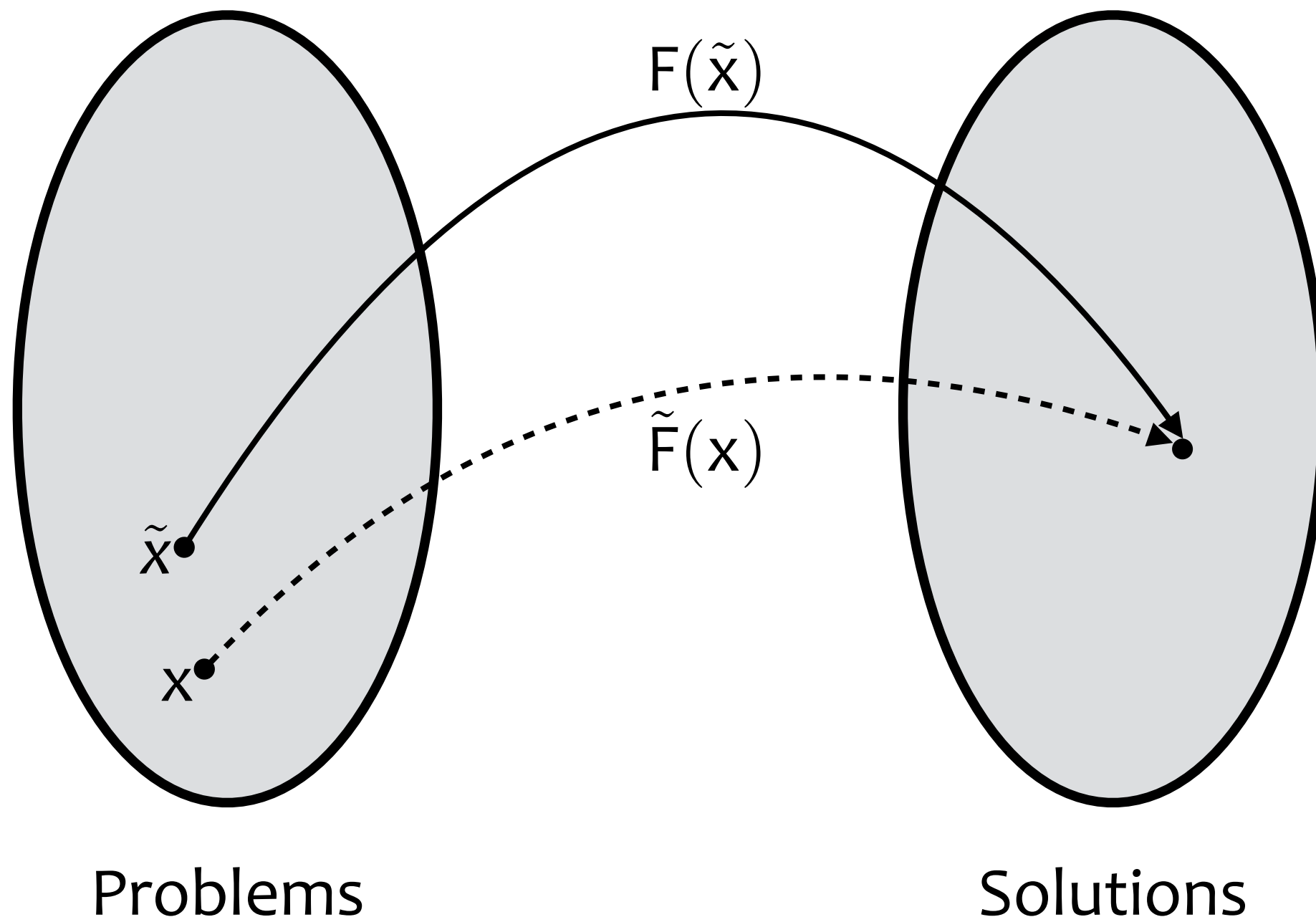
$$y + hf(y) + \frac{1}{2}h^2 F_f(\bullet)(y) + \frac{1}{8}h^3 F_f\left(\begin{smallmatrix} \bullet \\ \bullet \end{smallmatrix}\right)(y) + \left(\frac{1}{8} + \frac{1}{64\alpha_1} + \frac{1}{64\alpha_2}\right) h^3 F_f\left(\begin{smallmatrix} \bullet & \bullet \end{smallmatrix}\right)(y),$$

No way to correct this by
choosing α_1, α_2

Backward Error Analysis



Forward error analysis



Backward error analysis

Given a numerical solution map

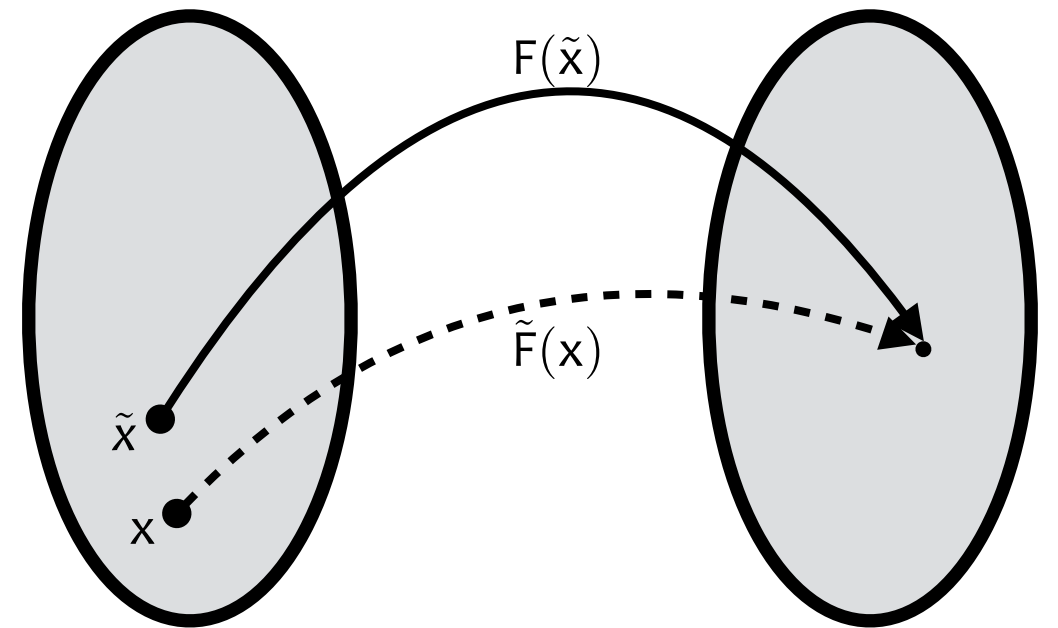
$$y_{n+1} = B(u, hf, y_n)$$

we want to find a modified flow

$$y'(t) = h^{-1} B(v, hf, y) = f_h(y)$$

such that the exact solution of the modified flow is the numerical solution of the original problem:

$$B(v \star e, hf, y) := B(e, B(v, hf, \cdot), y) = B(u, hf, y)$$

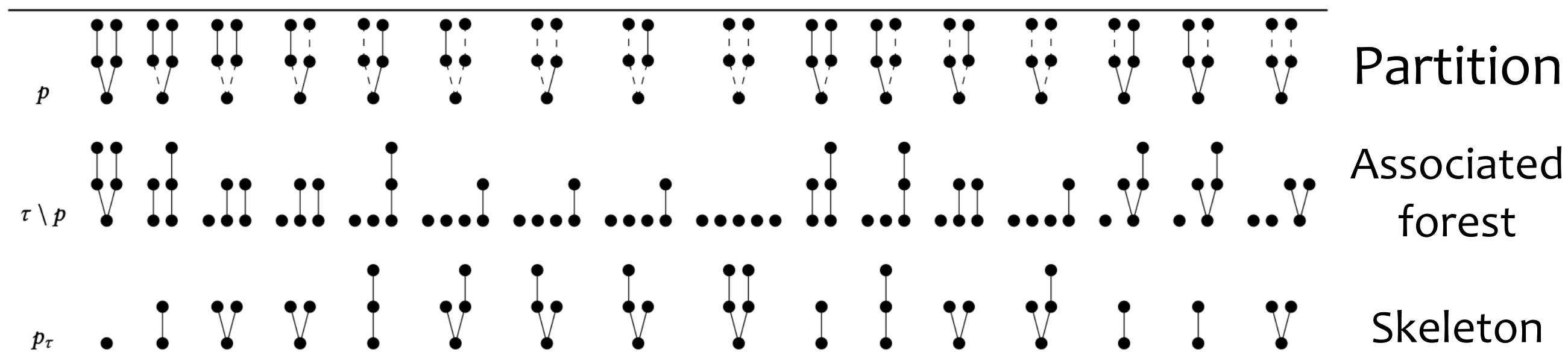


To find v , we just need to invert the substitution operation.

$$B(v \star u, hf, y) := B(u, B(v, hf, \cdot), y)$$

$$(v \star u)(\tau) = \sum_{p \in P(\tau)} v(\tau \setminus p) u(p_\tau)$$

See Chartier, Hairer, & Vilmart (2010)



Example:

2-stage, 2nd-order RK

$$\begin{array}{c|cc}
 0 & 0 & \\
 \frac{1}{2\alpha} & \frac{1}{2\alpha} & 0 \\
 \hline
 & 1 - \alpha & \alpha
 \end{array}$$

$$\begin{aligned}
 f_h = F_f(\bullet) &+ \frac{-h^2}{6} F_f\left(\begin{array}{c} \bullet \\ \bullet \end{array}\right) + h^2 \left(\frac{-1}{6} + \frac{1}{8\alpha} \right) F_f\left(\begin{array}{cc} \bullet & \bullet \\ \bullet & \bullet \end{array}\right) + \frac{h^3}{8} F_f\left(\begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array}\right) + h^3 \left(\frac{1}{8} - \frac{1}{16\alpha} \right) F_f\left(\begin{array}{cc} \bullet & \bullet \\ \bullet & \bullet \end{array}\right) \\
 &+ h^3 \left(\frac{1}{8} - \frac{1}{8\alpha} \right) F_f\left(\begin{array}{c} \bullet \\ \bullet & \bullet \end{array}\right) + h^3 \left(\frac{1}{24} - \frac{1}{16\alpha} + \frac{1}{48\alpha^2} \right) F_f\left(\begin{array}{ccc} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{array}\right) + O(h^4)
 \end{aligned}$$

$y'(t) = f_h(y)$ is referred to as the modified equation.

Example: Lotka-Volterra with explicit Euler

$$p'(t) = (2 - q)p$$

$$q'(t) = (p - 1)q$$

$$y_{n+1} = y_n + f(y)$$

Modified equation to first order:

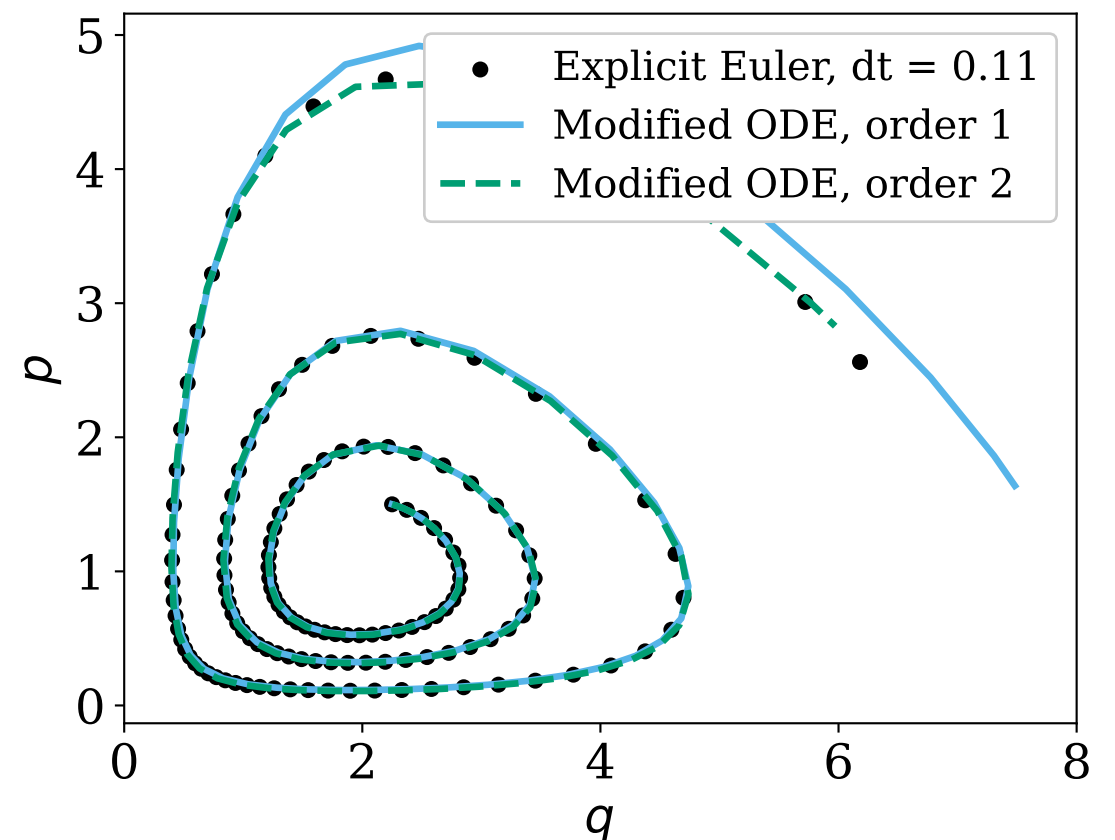
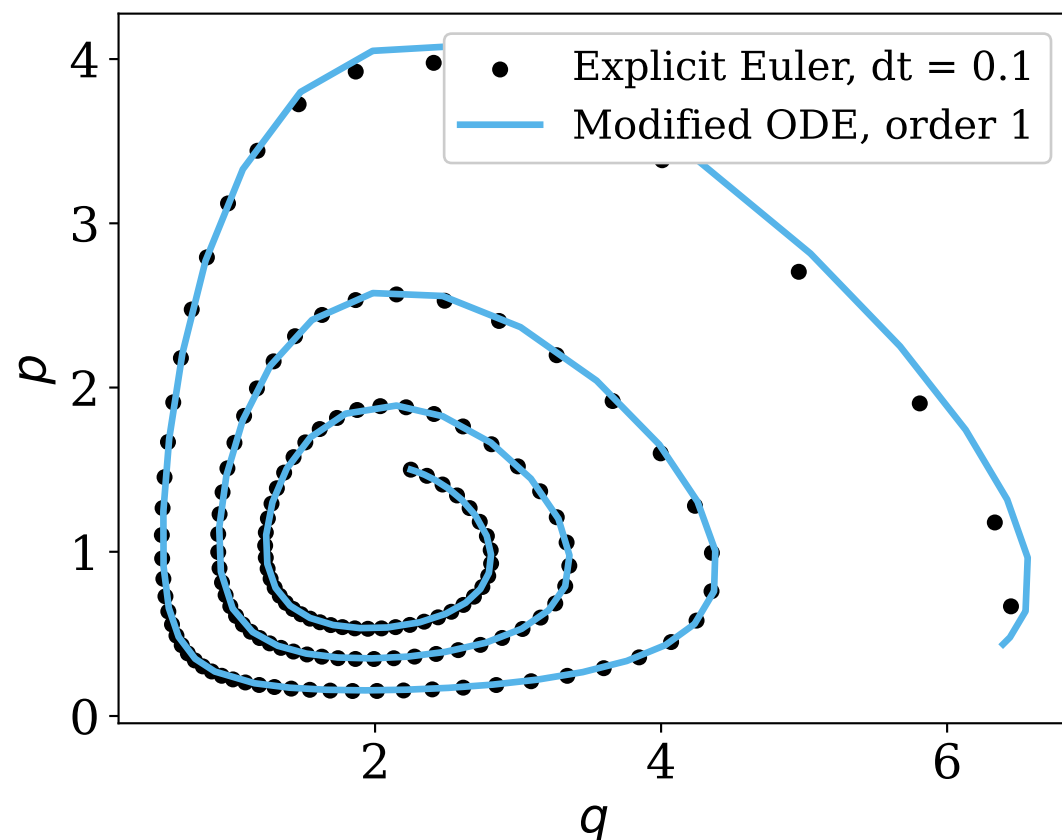
$$p'(t) = (2 - q)p + h \frac{((p - 1)q - (q - 2)^2)p}{2}$$

$$q'(t) = (p - 1)q + h \frac{((q - 2)p - (p - 1)^2)q}{2}$$

Example: Lotka-Volterra with explicit Euler

$$\begin{aligned}\dot{p}(t) &= (2 - q) \\ \dot{q}(t) &= (p - 1)\end{aligned}$$

$$y_{n+1} = y_n + f(y)$$

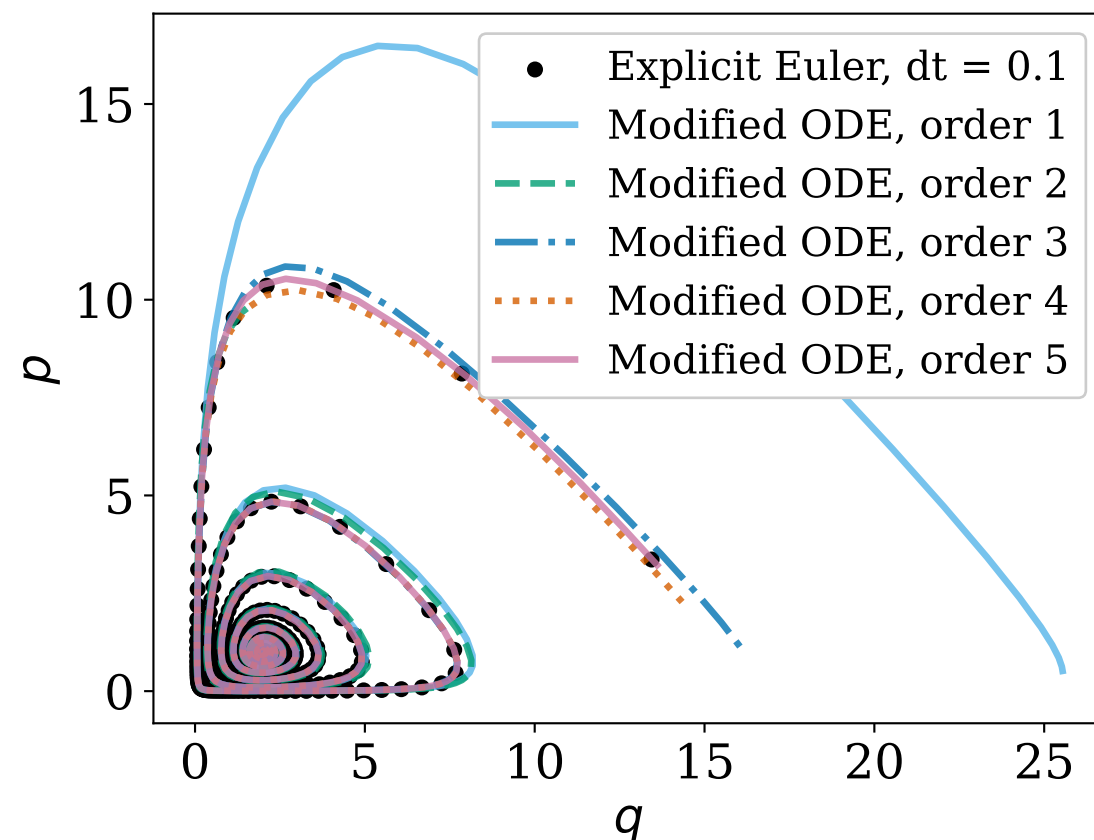


Example: Lotka-Volterra with explicit Euler

$$\dot{p}(t) = (2 - q)$$

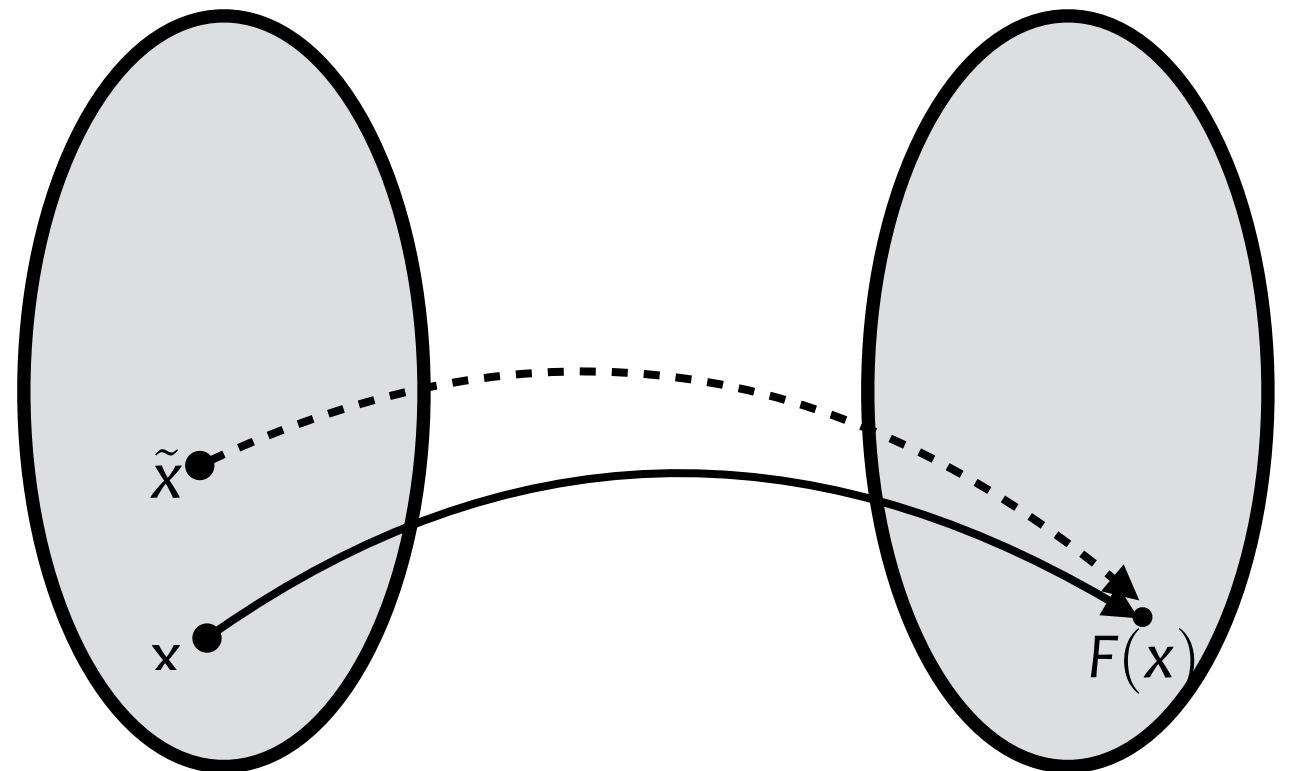
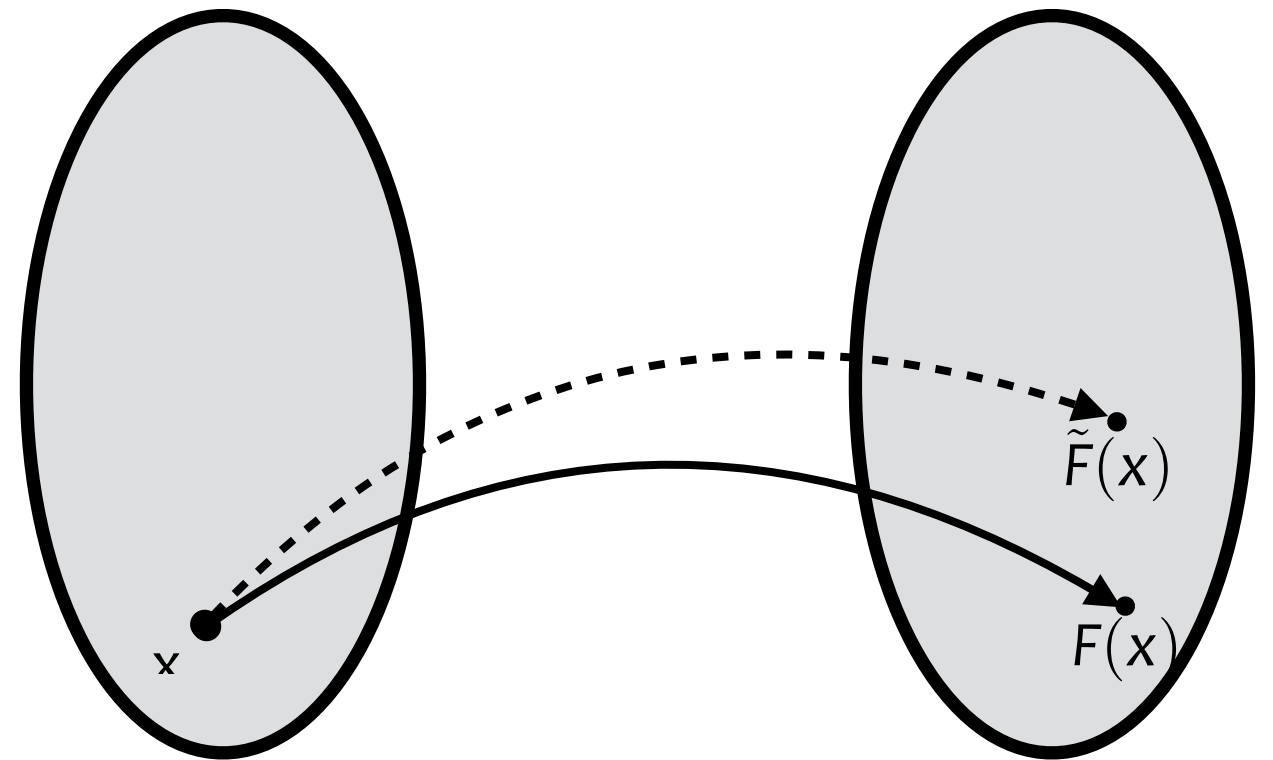
$$\dot{q}(t) = (p - 1)$$

$$y_{n+1} = y_n + f(y)$$



Modifying integrators

Idea: find a new differential equation, such that when we apply our numerical method to it, we get the exact solution of the original problem!



Lotka-Volterra + Euler

